Chapter 1

MATRICES AND DETERMINANTS

Learning Objectives

After studying this chapter, the students will be able to understand

- the definition of matrices and determinants
- the properties of determinants
- the concept of inverse matrix
- the concept of adjoint matrix
- the solving simultaneous linear equations
- the input-output analysis



Seki Kowa

1.1 Determinants

Introduction

The idea of a determinant was believed to be originated from a Japanese Mathematician Seki Kowa (1683) while systematizing the old Chinese method of solving simultaneous equations whose coefficients were represented by calculating bamboos or sticks. Later the German Mathematician Gottfried Wilhelm Von Leibnitz formally developed determinants. The present vertical notation was given in 1841 by Arthur Cayley. Determinant was invented independently by Crammer whose well known rule for solving simultaneous equations was published in 1750.

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In class X, we have studied matrices and algebra of matrices. We have also learnt that a system of algebraic equations can be expressed in the form of matrices. We know that the area of a triangle with vertices (x_1, y_1) (x_2, y_2) and (x_3, y_3) is

$$\frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

To minimize the difficulty in remembering this type of expression, Mathematicians developed the idea of representing the expression in determinant form and the above expression can be represented in the form

$$\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Thus a determinant is a particular type of expression written in a special concise form. Note that the quantities are arranged in the form of a square between

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